MSc THESIS DEFENSE · DTU WIND AND ENERGY SYSTEMS

Energy storage bids in day-ahead electricity markets

Can today's interval bids reproduce a state-of-charge-based market, what must a battery know to do it, and what does imperfect information cost?

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[Defense date] · working version, regenerated as results arrive

Six steps, each answering one question

THE PATH THROUGH THIS TALK

1	2	3	4	5	6
Motivation	Models	RQ1	RQ2	RQ3	Answers
Why storage bids matter	**Two ways to handle the SOC**	**Same outcome?**	**What must the battery know?**	**What forecast errors cost**	**Conclusions**
Why do storage bids matter now, and why not simply switch designs?	How do the two market designs treat the battery?	Can today's bids give Baldick's outcome?	What information do those bids need?	How much do forecast errors cost?	What does it mean for markets?

RQ1 Equivalence: can interval bids reproduce the SOC-based outcome? **RQ2** Information: what must the bids embed, and who can compute them? **RQ3** Cost: how much welfare is lost with imperfect price forecasts?

The blue steps carry the research questions; the others give the context needed to follow them.

PART 1 OF 6

Why storage bids matter



Why storage bids matter | Two ways to handle the SOC | Same outcome? | What must the battery know? | What forecast errors cost | Conclusions

Batteries are becoming price-setting resources

1 · WHY STORAGE BIDS MATTER

26 GW

US utility-scale batteries end 2024, +66% in one year

EIA, 2025

10 GW

batteries installed in ERCOT (Texas) by end 2024

Potomac Economics, 2025

>17 GW

storage in California by mid-2026

CAISO, 2026

2027

Europe's day-ahead market plans a new bid type for storage

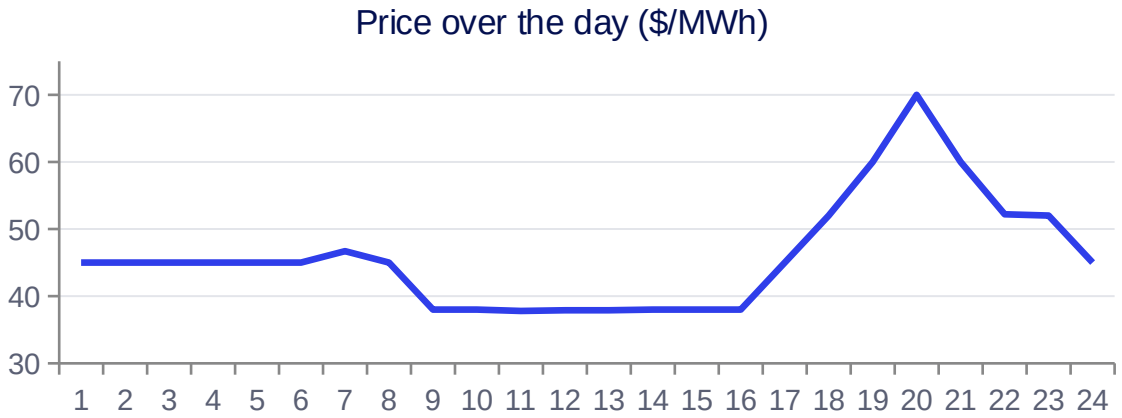
NEMO Committee, 2026

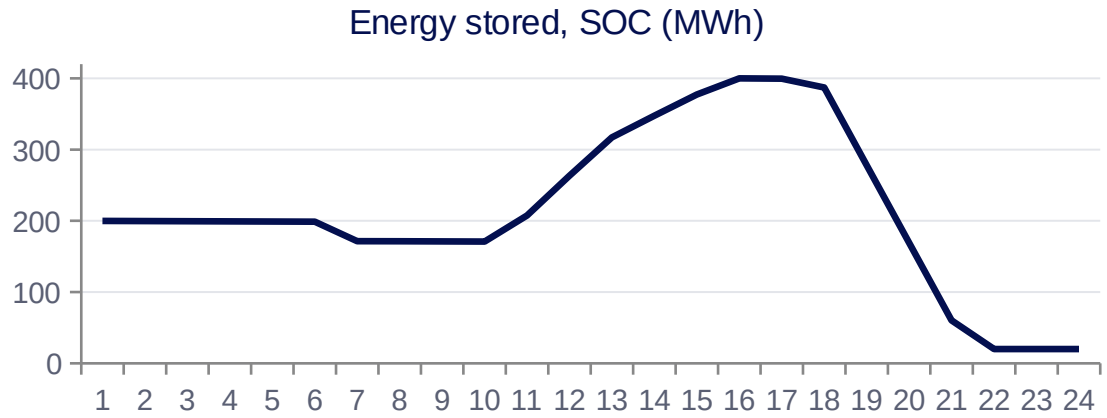
At this scale a battery no longer just takes the price: it often sets it. How it is allowed to bid changes prices and welfare for everyone.

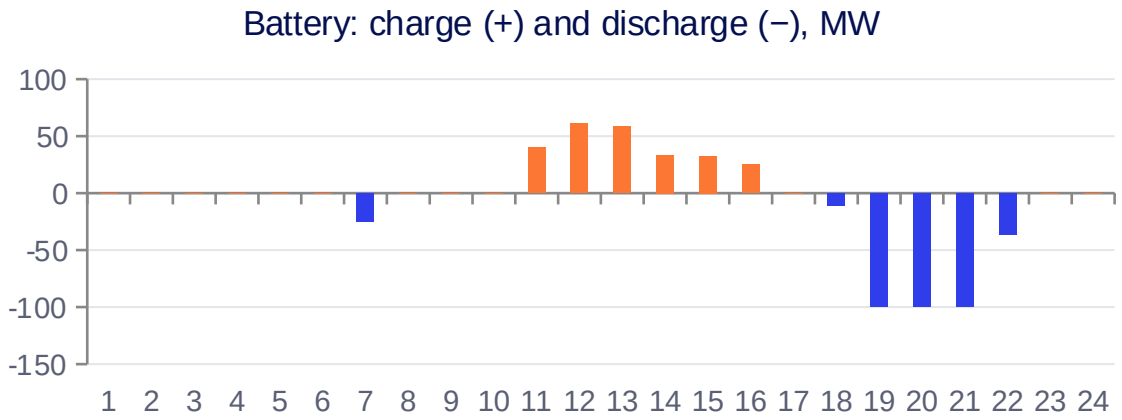
Storage market rules are no longer a detail: they decide prices in many hours.

A battery's value depends on its state of charge

1 · WHY STORAGE BIDS MATTER







Why it is different from a generator

A generator can sell in any hour independently. A battery can only sell energy it bought earlier, net of losses. Its decisions are linked across the whole day through the SOC.

Buy at midday, sell in the evening: the SOC links every hour of the day.

Who manages the SOC? Two market designs

1 · WHY STORAGE BIDS MATTER

TODAY

current design

The owner carries the SOC

Owner forecasts prices
and plans its SOC on its own

Bids every hour
a price to buy, a price to sell

ISO clears hour by hour
the SOC is not in the market

BALDICK

SOC-based design [1]

The market carries the SOC

Owner describes its battery
efficiencies, limits, wear

Parameters + one price w
w: value of energy left at the end

ISO optimises the SOC
together with the whole system

Real markets sit in between

- **Bids, no SOC** — ERCOT day-ahead, Europe
- **SOC + hourly bids** — CAISO, NYISO
- **SOC, no bids** — Baldick's proposal

The question is not whether batteries participate, but who carries the SOC: the owner or the market.

Why not simply adopt Baldick's design?

1 · WHY STORAGE BIDS MATTER

Reform cost and time

New rules, software, settlement and approval

Responsibility moves

The ISO operates assets it does not own

Model accuracy

Efficiency, degradation and hybrids are hard to describe

Strategy moves

Owners can misreport parameters and w

SOC deviations

Who pays when real time differs from day-ahead?

Computation

SOC rows couple all hours (next slides)

So we ask: can the current design reach Baldick's outcome through the right bids?

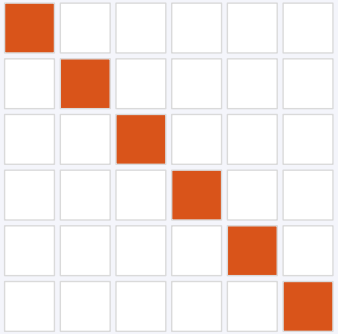
Baldick's model adds rows that couple every hour of the day

Extra size of Baldick's model (E batteries, T intervals)

$$\Delta\text{rows} = 3ET \qquad \Delta\text{columns} = ET \qquad \Delta\text{nonzeros} = 8ET - 3E$$

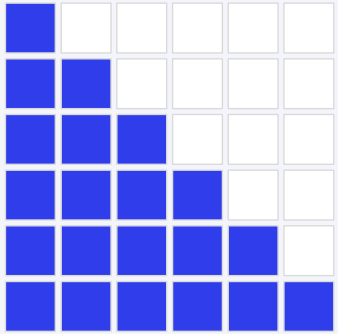
Structure matters more than size

Today



each hour on its own: can be split by hour

Baldick



the SOC links each hour to every hour before it

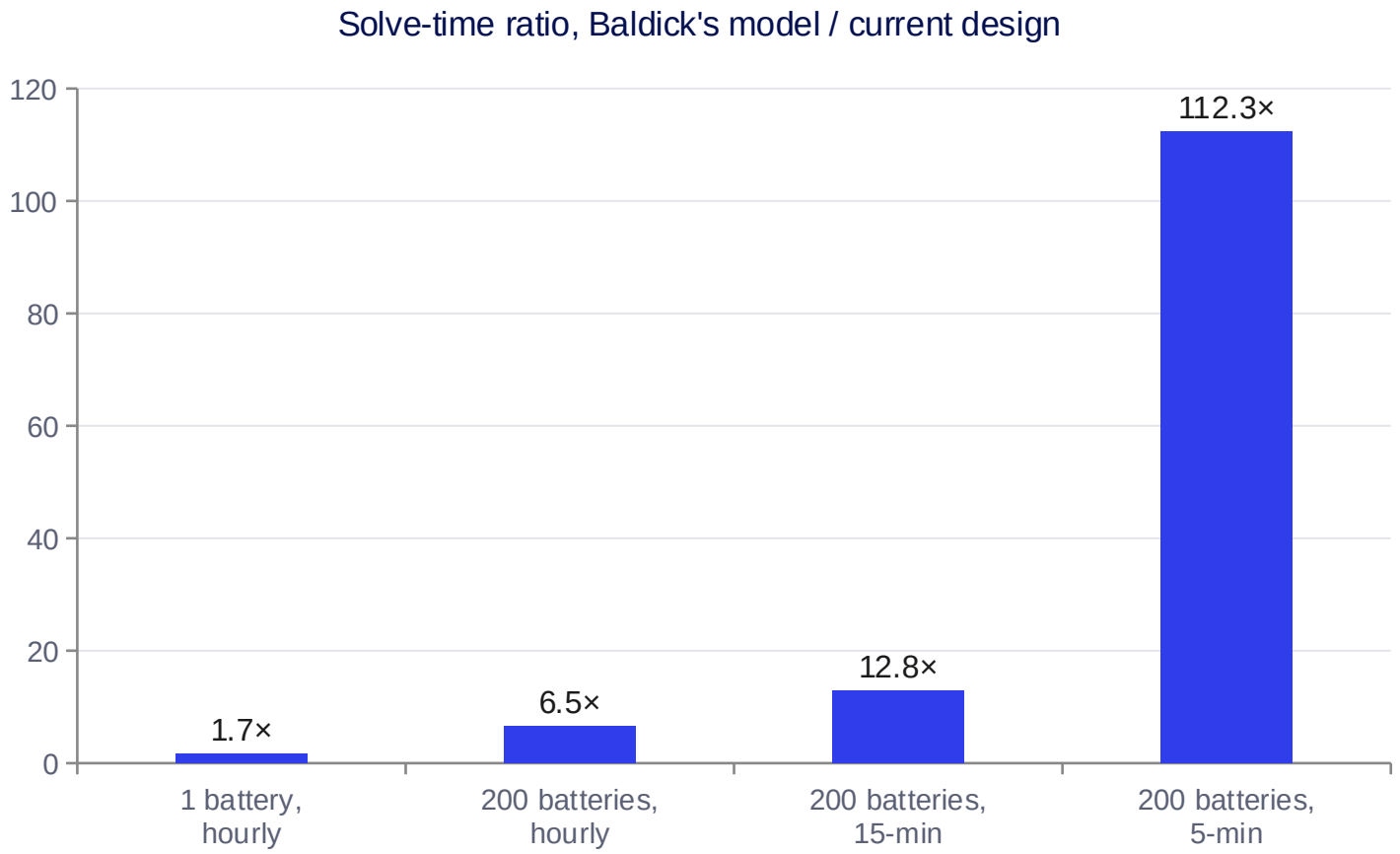
rows: hour t · columns: hours whose decisions enter hour t's constraints

Still a linear program

No binary variables are needed to stop the battery charging and discharging at once: round-trip losses already make it unprofitable whenever stored energy has a non-negative value.

Size grows linearly; the coupling across hours is what makes it heavier.

The gap grows with batteries and time resolution



44.7 s vs 0.4 s

200 batteries, 5-minute resolution, one day

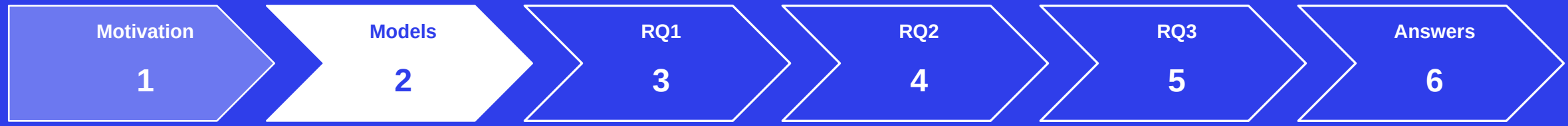
How we use it

The burden matters much less after the market has cleared. Baldick's model is an ideal ex post benchmark: run it afterwards to measure what the actual market lost.

Baldick's model is heavier in operation, but ideal as a benchmark after the fact.

PART 2 OF 6

Two ways to handle the SOC



Why storage bids matter | **Two ways to handle the SOC** | Same outcome? | What must the battery know? | What forecast errors cost | Conclusions

What the symbols mean

2 · TWO WAYS TO HANDLE THE SOC

Battery parameters θ (what the owner submits to Baldick)

α	charging efficiency: MWh stored per MWh bought
β	MWh taken from storage per MWh sold (β > α: losses)
γ	share of stored energy kept each hour
ε, ζ	wear cost per MWh charged / discharged
$\underline{s}, \bar{s}$	minimum / maximum energy stored
$\bar{p}, \bar{g}$	maximum charging / discharging power

w willingness to pay for each MWh still stored at the end of the day: how much the owner values keeping energy for tomorrow ($/MWh)

Market, bids and what the model computes

λt	price at the battery's location in hour t
ut, ct	bid to buy (charge) / offer to sell (discharge)
πt	value of one MWh stored at the end of hour t
$\bar{\nu} t, \underline{\nu} t$	extra value when the battery is full / empty
pt, gt, st	charging, discharging, energy stored
ΔT	length of an interval (hours)

π, the value of stored energy, is the key quantity of the whole talk.

The battery in Baldick's model

2 · TWO WAYS TO HANDLE THE SOC



SOC dynamics (multiplier π: value of stored energy)

$$s_t = \underbrace{\gamma\, s_{t-1}}_{\text{kept}} + \underbrace{\Delta T\, \alpha\, p_t}_{\text{charged}} - \underbrace{\Delta T\, \beta\, g_t}_{\text{discharged}}$$

SOC limits within the hour, and power limits (multipliers ν̄, ν: extra value when full / empty)

$$\gamma s_{t-1} + \Delta T \alpha p_t \le \bar{s} \qquad \gamma s_{t-1} - \Delta T \beta g_t \ge \underline{s} \qquad 0 \le p_t \le \bar{p}, \quad 0 \le g_t \le \bar{g}$$

Three numbers describe the losses (α, β, γ); one number, w, values the energy kept for tomorrow.

Two clearing problems, one difference: who prices stored energy

Current design: the battery enters with its bids

$$\min \sum_t \Delta T \Big[\underbrace{\text{generation cost} - \text{demand value}}_{\text{rest of the market}} + \underbrace{c_t\, g_t - u_t\, p_t}_{\text{battery: its bids}} \Big]$$

Baldick's design: the battery enters with its parameters, + the SOC rows of the previous slide

$$\min \sum_t \Delta T \Big[\underbrace{\text{generation cost} - \text{demand value}}_{\text{rest of the market}} + \underbrace{\epsilon\, p_t + \zeta\, g_t}_{\text{battery: wear}} \Big] - \underbrace{w \left(s_T - \bar{\bar{s}}_T \right)}_{\text{value of energy kept}}$$

Everything else (network, generators, demand) is identical, so any difference in outcome comes from the battery.

Baldick's design is a market equilibrium, not only an optimum

2 · TWO WAYS TO HANDLE THE SOC

Operator's problem
maximise welfare

split by owner

Generator

sells at its local price

Load

buys at its local price

Battery owner

its own profit problem

Network

buys at one end, sells at the other

Market clearing
balance at every bus
(the only condition owned by nobody)

Theorem (equilibrium)

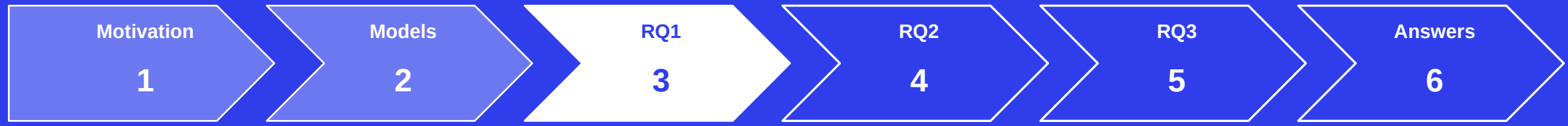
Under price-taking, a convex market, separability by owner and a non-increasing w, the operator's optimum is a competitive equilibrium: every participant maximises its own profit at the prices.

Used later: the battery's block is exactly its own profit-maximisation problem, so π and ν belong to the owner.

Baldick's outcome is one every participant would choose itself at those prices.

PART 3 OF 6

Same outcome?

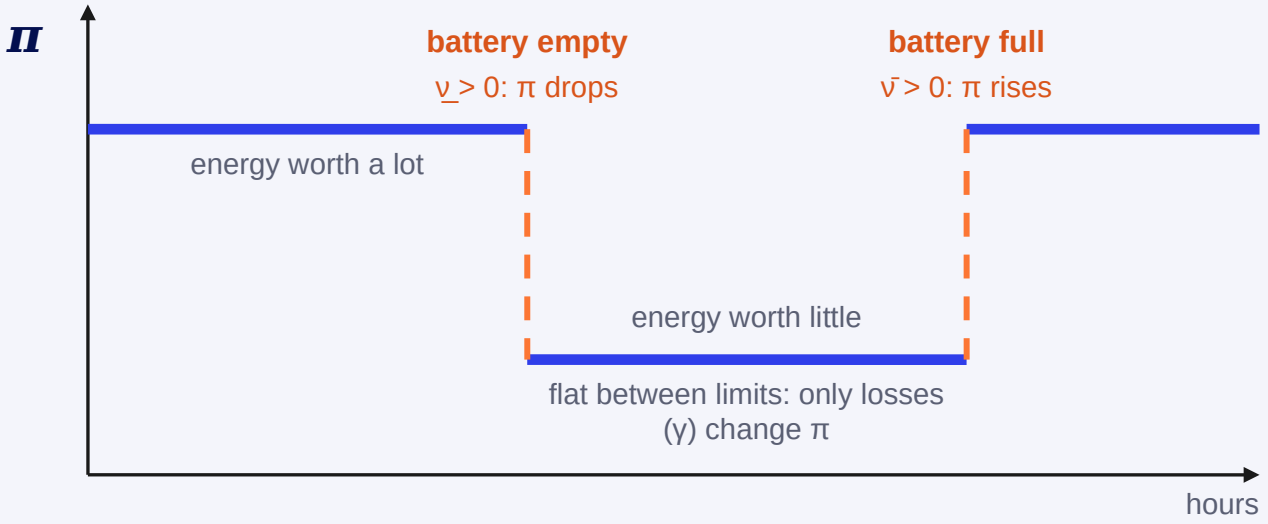


Why storage bids matter | Two ways to handle the SOC | **Same outcome?** | What must the battery know? | What forecast errors cost | Conclusions

First, the multipliers: π is the value of stored energy

3 · SAME OUTCOME?

How π moves over a day (sketch)



π value of stored energy

what one more MWh in the battery is worth ($/MWh)

$\bar{\nu}$ extra value when full

what one more MWh of room would be worth

$\underline{\nu}$ extra value when empty

what one more MWh of energy would be worth

$$\pi_t = \gamma\left(\pi_{t+1} - \bar{\nu}_{t+1} + \underline{\nu}_{t+1}\right), \qquad \pi_T = w$$

Read backwards from the end of the day: π starts at w, is carried back through the losses γ, and jumps only where the battery is full or empty.

π and ν put a price on stored energy: with them, the battery's choice in each hour becomes simple.

With π and ν, both markets give the battery two price thresholds per hour

3 · SAME OUTCOME?

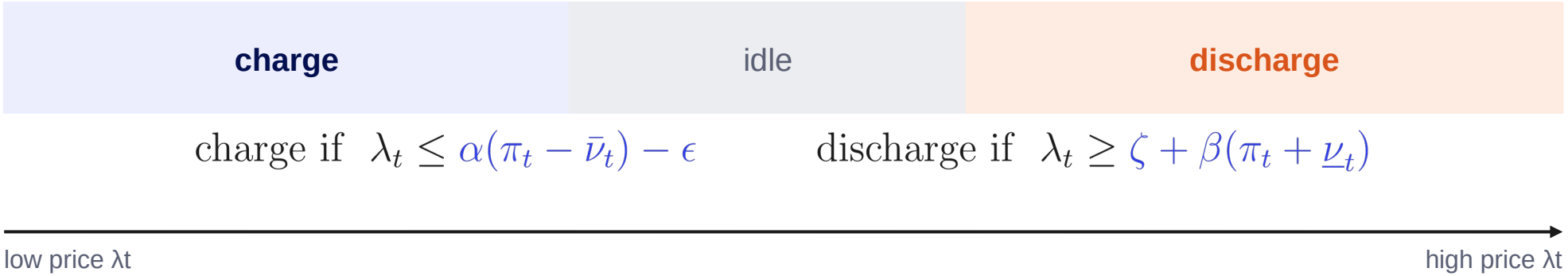
Today

thresholds = its bids



Baldick

thresholds from π, ν



Charge threshold: α·π − ε

A MWh bought becomes α MWh stored, worth α·π, minus wear.

Discharge threshold: ζ + β·π

A MWh sold uses β MWh of storage, worth β·π, plus wear.

If the two pairs of thresholds coincide in every hour, the battery behaves identically in both markets.

Equivalent bids: bid exactly at Baldick's thresholds

3 · SAME OUTCOME?

Equivalent bids (E)

$$u_t = \alpha\left(\pi_t - \bar{\nu}_t\right) - \epsilon \qquad c_t = \zeta + \beta\left(\pi_t + \underline{\nu}_t\right)$$

1 Same rest of the market

Network, generators and demand enter both problems identically.

2 Same battery thresholds

With (E), the bid and offer are Baldick's thresholds in every hour.

3 Same optimality conditions

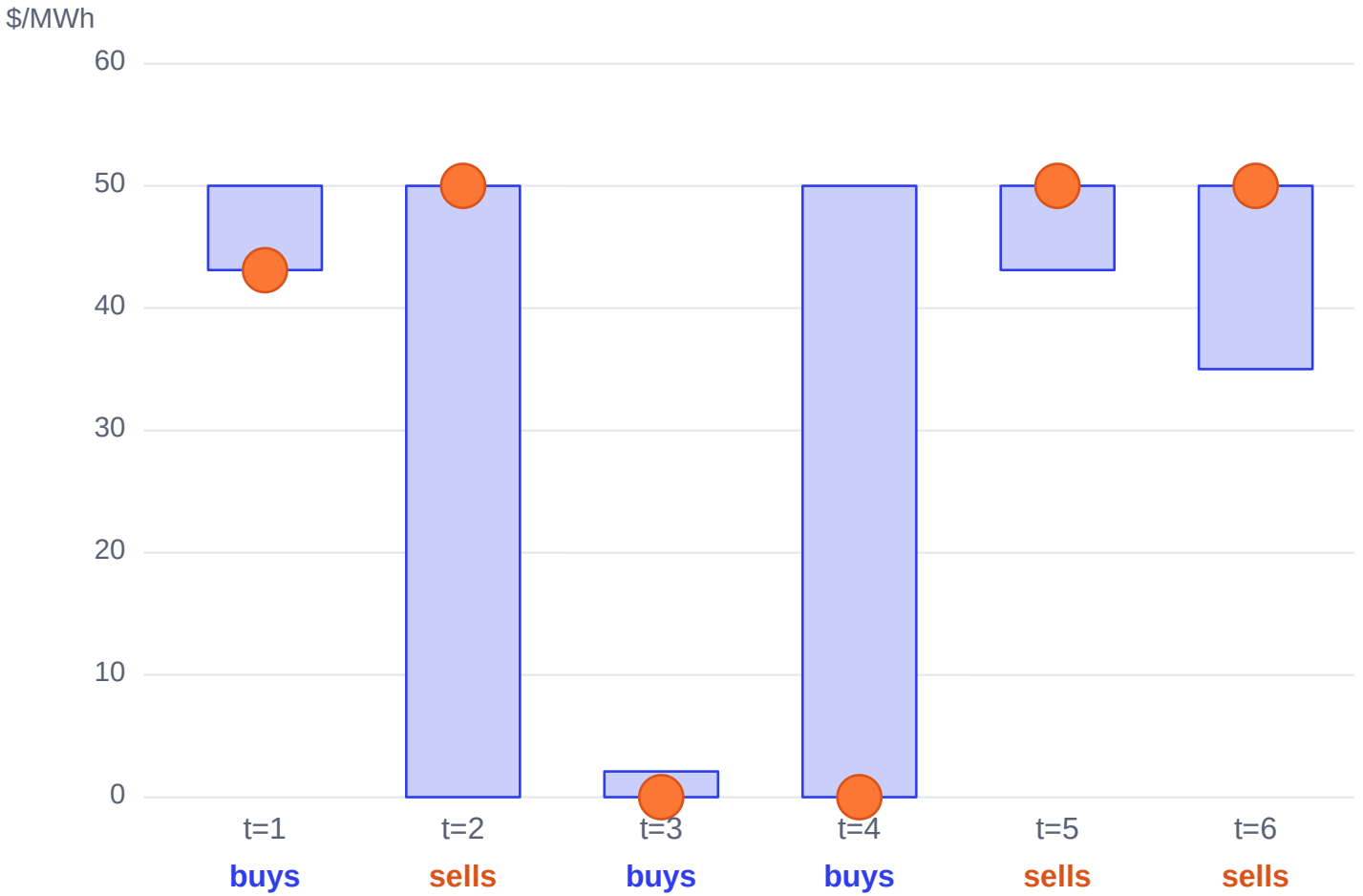
Both problems are linear, so the same conditions give the same optimum.

Result (Proposition 1): with bids (E), the current design clears at Baldick's dispatch and Baldick's prices, for any network, ramps and number of batteries.

Yes: today's design can reproduce Baldick's outcome. The question becomes what the bids require.

In the example: the price always sits on an edge of the bid band

3 · SAME OUTCOME?



Baldick's example [1]

One bus, six 4-hour intervals, cheap renewables and a 50 $/MWh thermal unit. One 800 MWh battery, w = 40 $/MWh.

How to read it

- band from bid u (bottom) to offer c (top)
- price λ* of Baldick's design

Wide bands at t = 2 and 4

The battery is empty or full there: π jumps.

The bids reproduce Baldick's prices in every interval of the example.

The catch: the bids use numbers that only exist inside Baldick's market

3 · SAME OUTCOME?

$$u_t = \alpha\left(\pi_t - \bar{\nu}_t\right) - \epsilon \qquad c_t = \zeta + \beta\left(\pi_t + \underline{\nu}_t\right)$$

π and ν are multipliers of Baldick's optimisation. In today's market nobody solves that problem, so nobody has them.

What they contain (Corollary 1)

Private data of the battery (α, β, γ, ε, ζ, w) and its own SOC rents, which depend on everything else in the market…

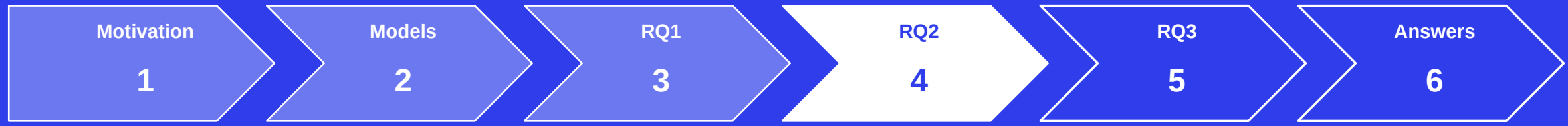
…but only through its local prices

Part 4 shows that the rents depend on the rest of the market only through the price path at the battery's own bus.

RQ1 is answered with Baldick's multipliers; RQ2 asks whether the owner can do without them.

PART 4 OF 6

What must the battery know?



Why storage bids matter | Two ways to handle the SOC | Same outcome? | **What must the battery know?** | What forecast errors cost | Conclusions

The battery's own problem gives the right numbers

The battery's own profit-maximisation problem at prices λ

$$\max_{p,g,s} \sum_{t} \Delta T \left[\lambda_t \left(g_t - p_t\right) - \epsilon\, p_t - \zeta\, g_t\right] + w\left(s_T - \bar{\bar{s}}_T\right) \quad \text{s.t. the battery's own limits}$$

Inputs

parameters θ
+ local prices λ

Own problem

small LP, one battery

Any optimal dual

π, ν̄, ν̲

Bids (E)

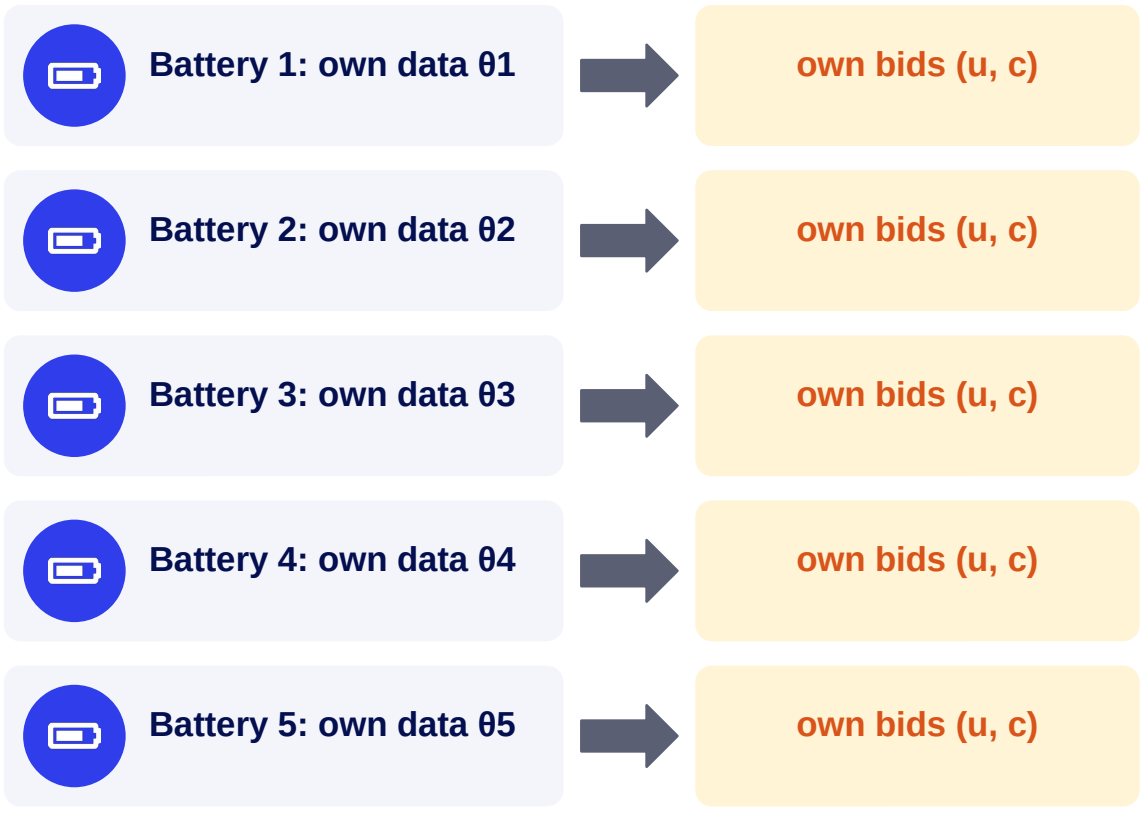
u, c for every hour

Theorem 1. Take any optimal dual of each battery's own problem at its local prices λ*, and submit (E). Then the current design clears at Baldick's dispatch and prices. The bids depend only on the battery's parameters and its local prices: not on other participants, the network, ramps, or the multipliers the operator's solver happens to return.

The gap between the two designs is a gap in price information, not in market design.

It works for any number of batteries, each on its own

4 · WHAT MUST THE BATTERY KNOW?



Today's market clears all bids

200

random joint choices of duals

0

optimality violation (machine precision)

0

price difference to Baldick (machine precision)

Given the prices, batteries need no coordination: each one builds its bids alone.

Between two limits, stored energy has one value: segments



Inside a segment

no limit binds: a MWh can be moved freely between its hours, so it has one value π (only losses γ change it).

At a limit

the battery is full or empty: π jumps, and the jump is ν.

So

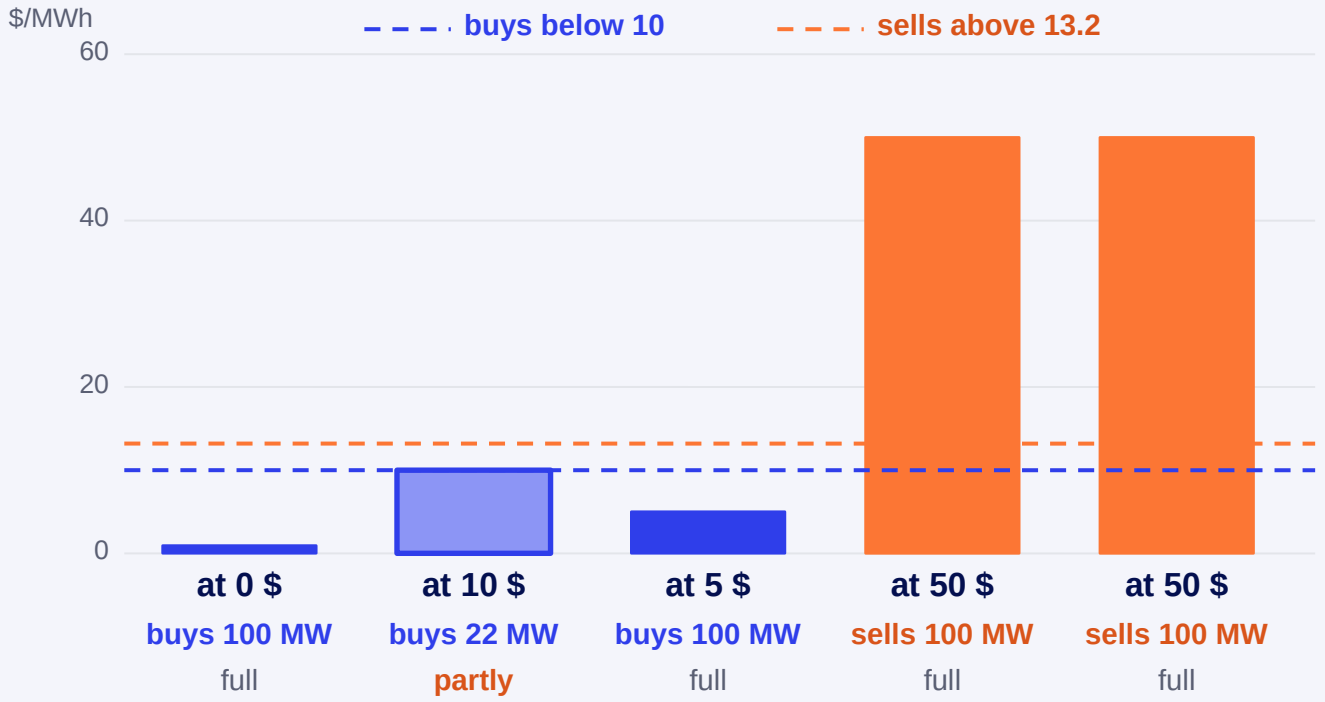
the bids of a whole segment hang on one number. Which price sets it? Next slide.

The battery fills and empties; between those moments its energy has a single value π.

Inside a segment, the last MWh the battery buys or sells sets π

4 · WHAT MUST THE BATTERY KNOW?

One segment, five hours: price and what the battery does



1. It can sell 200 MWh at 50, so it needs 222 MWh bought (losses).
2. It buys the cheapest hours first: 0 and 5 in full, then only 22 MW at 10.
3. That last MWh, bought at 10, is the marginal one: storing it is worth exactly its price. So π = 12.2.

Which prices matter

In each segment: the highest price at which it charges only partly, or the lowest at which it discharges only partly.

$$\pi_\tau = \frac{\lambda_\tau + \epsilon}{\alpha} \quad \text{(partial charge)} \qquad \pi_\tau = \frac{\lambda_\tau - \zeta}{\beta} \quad \text{(partial discharge)}$$

A price error in the marginal hour moves the bids of its whole segment.

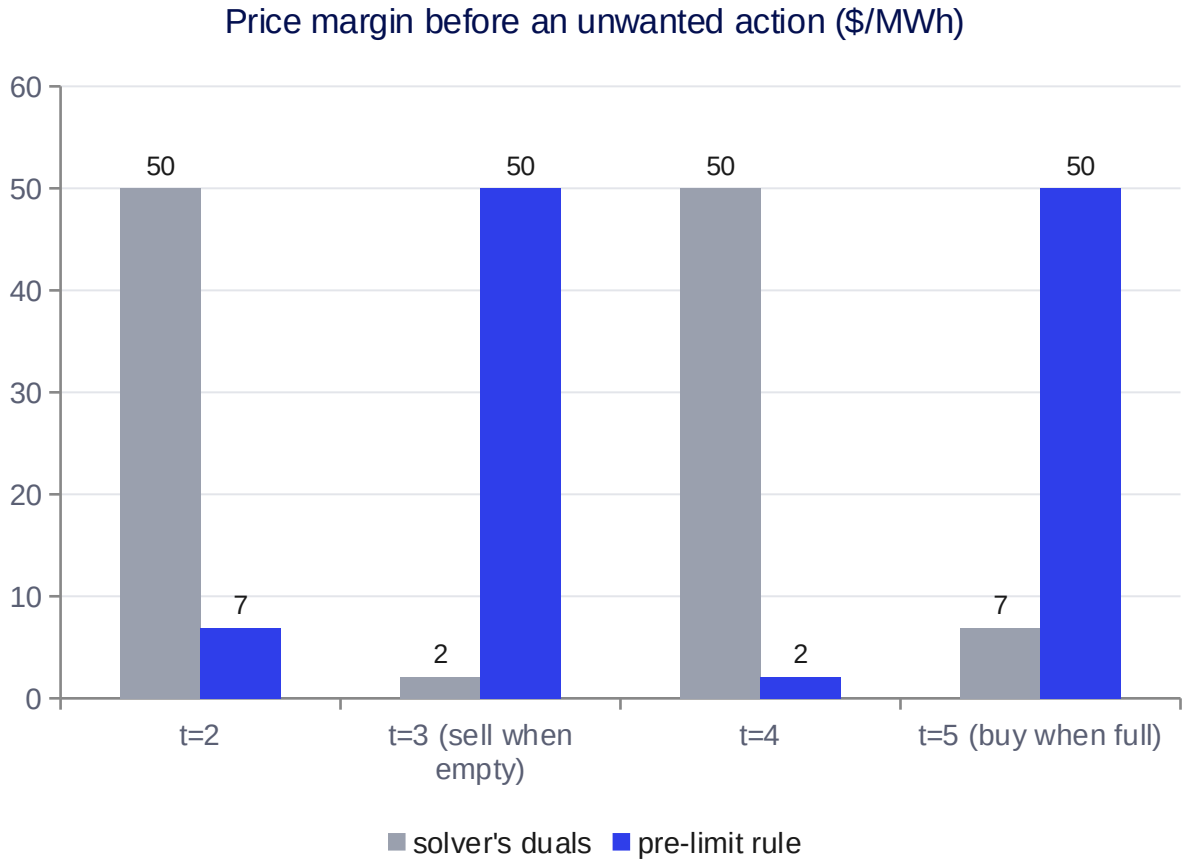
When the battery is full or empty, prices give a range: choose robustly

$$\pi_t^R = \min\left\{ \max\left\{ \frac{\pi_{t-1}^R}{\gamma},\ \pi_t^- \right\},\ \pi_t^+ \right\}$$

Pre-limit rule: keep the value carried in from the previous hour whenever it is admissible.

Why it matters

Selling when empty or buying when full is physically impossible: a small price error there makes the schedule infeasible. The pre-limit rule gives those hours a 50 $/MWh margin, and accepts small margins only where the unwanted action is feasible.



Non-uniqueness is not a flaw: it is a free choice we use for robustness.

Answer to RQ2: what the bids embed, and who can compute them

4 · WHAT MUST THE BATTERY KNOW?

Information	Examples	Who has it
Private to the battery	α, β, γ, ε, ζ, limits, w	the owner (submitted in Baldick's design)
Local prices	prices at the battery's bus, above all in its marginal hours	**nobody in advance: must be forecast**
SOC path	hours where it becomes full or empty	follows from the own problem at those prices

No SOC limit binds

Technical data alone are enough: u = αγ^(T−t)w − ε, c = ζ + βγ^(T−t)w.

Market enforces the SOC

With SOC rows in the clearing (CAISO, NYISO), technical-data bids are always exact.

Two users of the same map

The owner, with its own forecast, to bid. The operator, with its forecast, to build reference bids for market monitoring.

With perfect local prices the gap closes; with imperfect ones it does not. That is part 5.

Every result is verified numerically on Baldick's three cases

4 · WHAT MUST THE BATTERY KNOW?

Residual in each case (0 = machine precision)

Check	Result	Base	w = 55	$\bar{s}$ = 1200
Same prices with bids (E)	Prop. 1	0	0	0
Optimal values differ by the constant K	Prop. 2	0	0	0
π rebuilt from prices only	Prop. 4	0	0	0
Own problem = Baldick's battery block	Lemma	0	0	0
Range of π: polyhedron = dual face	Prop. 5	0	0	0
Any admissible dual reproduces the outcome	Thm. 1	0	0	0

8

checks

3

cases

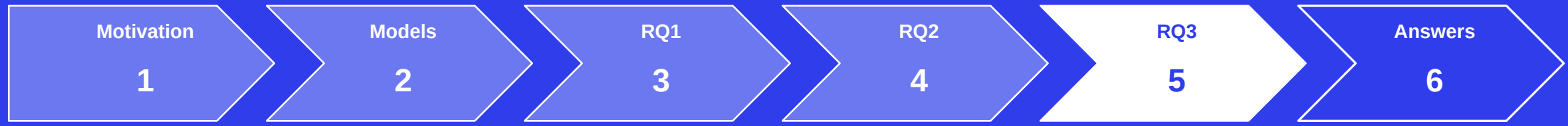
0

residual in every check (machine precision)

The theory holds to machine precision on every published case.

PART 5 OF 6

What forecast errors cost



Why storage bids matter | Two ways to handle the SOC | Same outcome? | What must the battery know? | **What forecast errors cost** | Conclusions

Four ways a battery can bid, from full information to none

5 · WHAT FORECAST ERRORS COST

B★	true prices + Baldick's quantities	upper bound: loss = 0
B★p	true prices + its own plan	isolates the quantity loss
Bfcst	a price forecast $\hat{\lambda}$ + its own plan	main experiment
Bpriv	technical data only, no forecast	lower bound

less information

Quantity rule

When the battery sets the price, the market is indifferent to how much it trades. Each battery bids its own planned quantity just inside its bid (± δ = 0.01 $/MWh) and the rest of its capacity just outside, so ties are broken in favour of its plan.

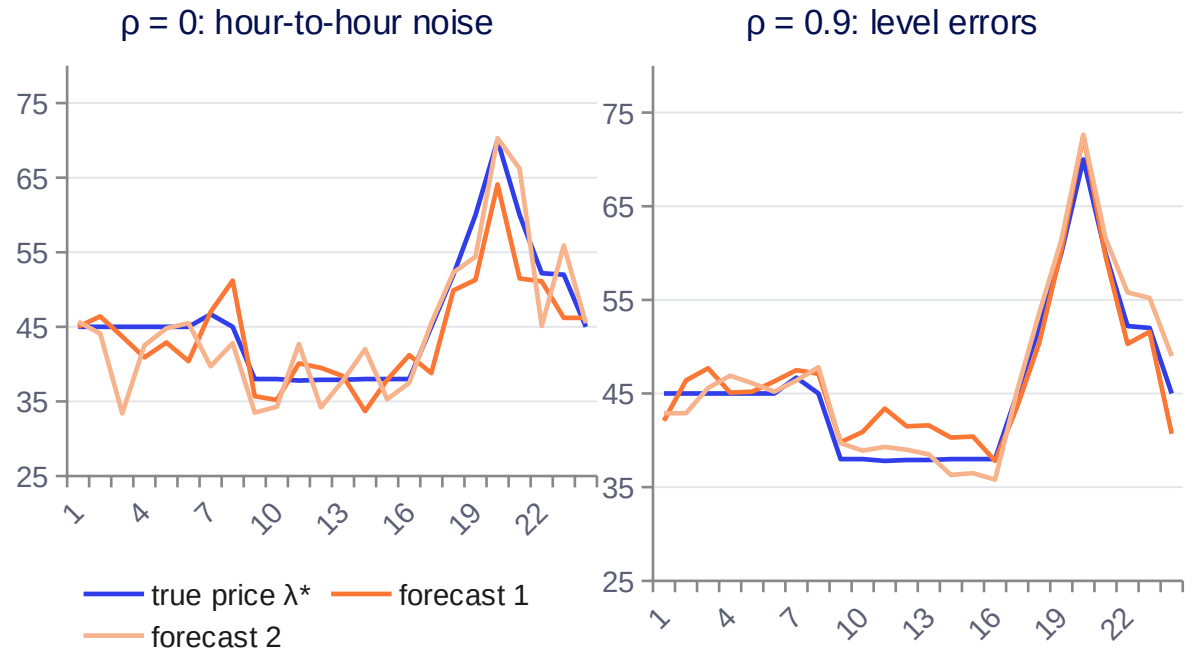
B★ and Bpriv bracket every result; the forecast is useful only between them.

How we perturb the forecast: size, persistence, bias, common share

$$\hat{\lambda}_{e,t} = \lambda^{\star}_{t} + \underbrace{b\,\bar{\lambda}}_{\text{bias}} + \underbrace{\sigma\,\bar{\lambda}\,z_{e,t}}_{\text{noise}}$$

$$z_{e,t} = \sqrt{\omega}\,\chi_t + \sqrt{1-\omega}\,\eta_{e,t} \qquad x_t = \rho\,x_{t-1} + \sqrt{1-\rho^2}\,w_t$$

Parameter	Controls	Values
σ	size	0–40% of mean price
ρ	persistence	0, 0.5, 0.9
b	bias	−10, 0, +10%
ω	common share	0, 0.5, 1



Example: σ = 10% of the mean price. Same average error, very different consequences.

Noise changes when the battery trades; persistent errors change how much the energy seems worth.

One Monte Carlo draw, step by step

5 · WHAT FORECAST ERRORS COST

1 · Benchmark

Solve Baldick once: λ*, schedule, welfare SW*

2 · Forecast

Draw errors, build $\hat{\lambda}$ for each battery

3 · Bids

Own problem at $\hat{\lambda}$ → pre-limit rule → bids (E) + quantity rule

4 · Day-ahead

ISO clears today's market with the true system

5 · Real time

Battery delivers the closest feasible schedule; others re-dispatch at offer + ψ

6 · Evaluate

Welfare loss, undeliverable energy, prices

Steps 2–6 are repeated hundreds of times per setting; the benchmark is solved once.

What we measure: welfare lost, as a share of what storage is worth

5 · WHAT FORECAST ERRORS COST

$$\Delta SW = SW^{\star} - SW^{\text{real}} \geq 0 \qquad \text{loss} = \frac{\Delta SW}{SW^{\star} - SW^{\text{no storage}}} \ (\% \text{ of storage value})$$

$$\Delta SW = \underbrace{\Delta SW\left(\mathcal{B}^{\star p}\right)}_{\text{quantity information}} + \underbrace{\Delta SW\left(\mathcal{B}^{\text{fcst}}\right) - \Delta SW\left(\mathcal{B}^{\star p}\right)}_{\text{price forecast}}$$

Undeliverable energy

MWh the battery cannot deliver

Price distortion

mean |λCD − λ*|

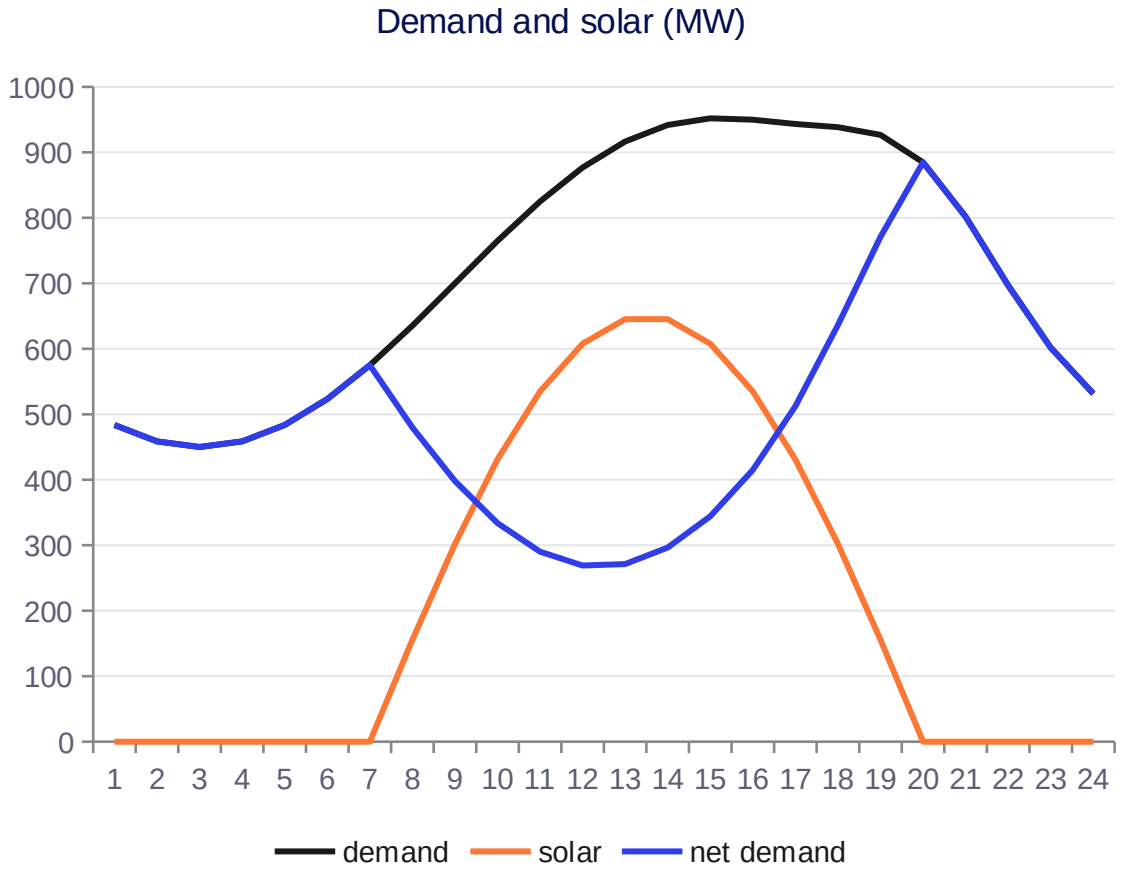
Value of a forecast

σ* where Bfcst and Bpriv lose the same

Normalising by the storage value makes systems, sizes and days comparable.

Pilot: a 24-hour system with one battery

5 · WHAT FORECAST ERRORS COST



100 MW / 400 MWh

battery, 4 hours, ~85% round trip

45.9 $/MWh

mean benchmark price (38 midday, 70 peak)

7,867 $/day

value storage creates in the benchmark

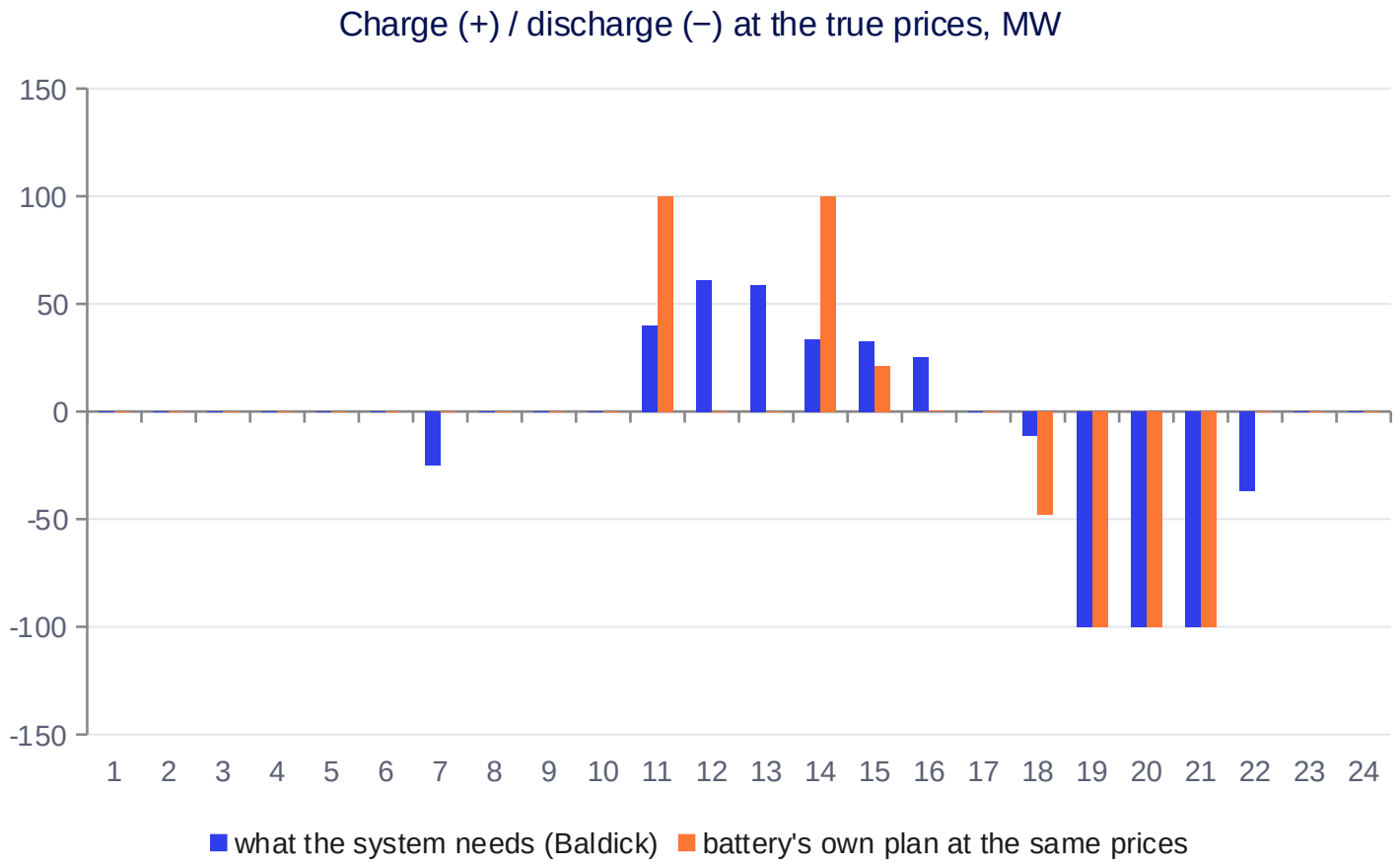
150 draws

per setting in the pilot (500 planned)

Small battery, realistic day: the battery sets the price only in some hours.

Result 1: perfect prices are not enough — quantities matter

5 · WHAT FORECAST ERRORS COST



11.1%

of the storage value lost with perfect prices (875 $/day)

0%

with Baldick's quantities (B★): confirms the theory

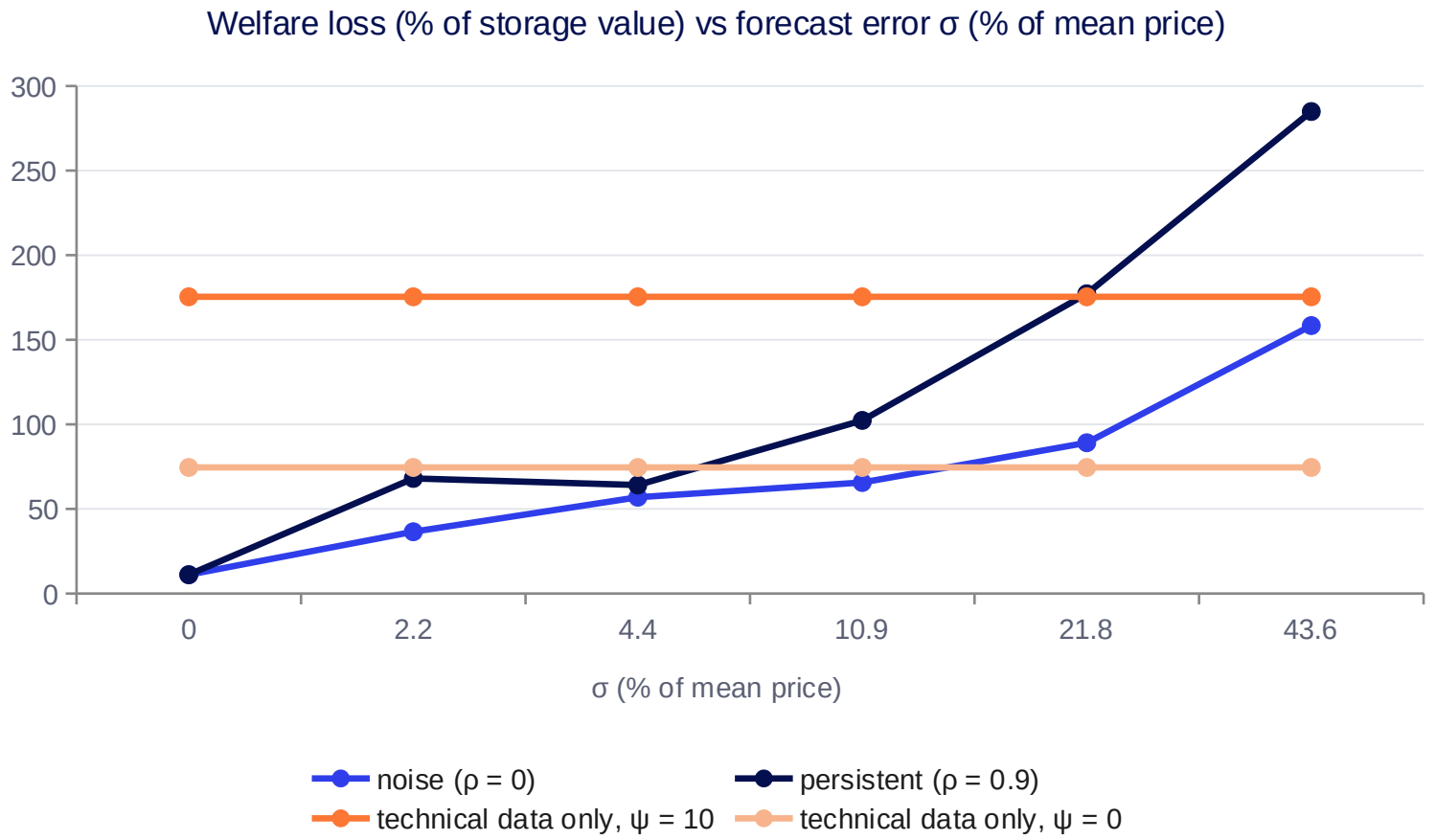
Why

Flat prices make many schedules equally profitable to the battery, but not equally good for the system.

Interval bids carry prices, not quantities: part of the gap is quantity information.

Result 2: small errors already cost a lot, persistent ones most

5 · WHAT FORECAST ERRORS COST



36.5%

noise of 2% of the mean price

65.6% / 102.3%

at 11%: noise / persistent

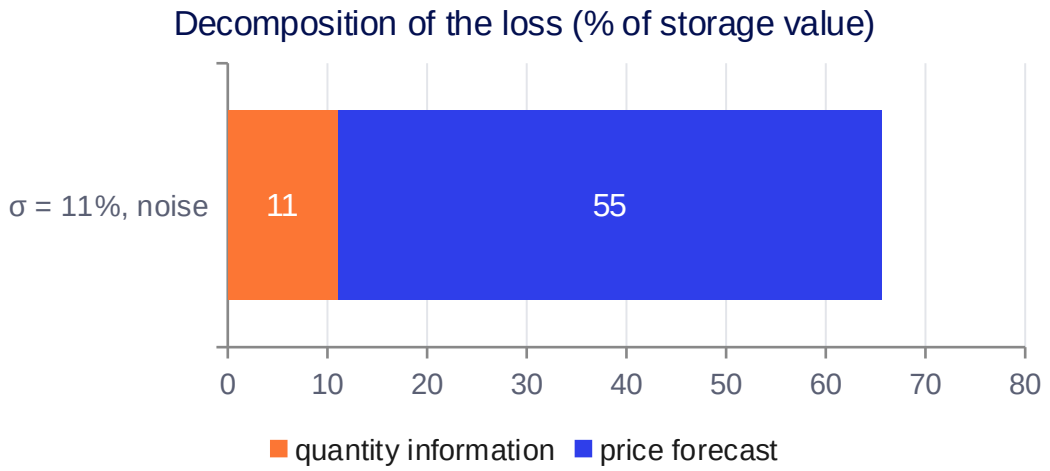
>100%

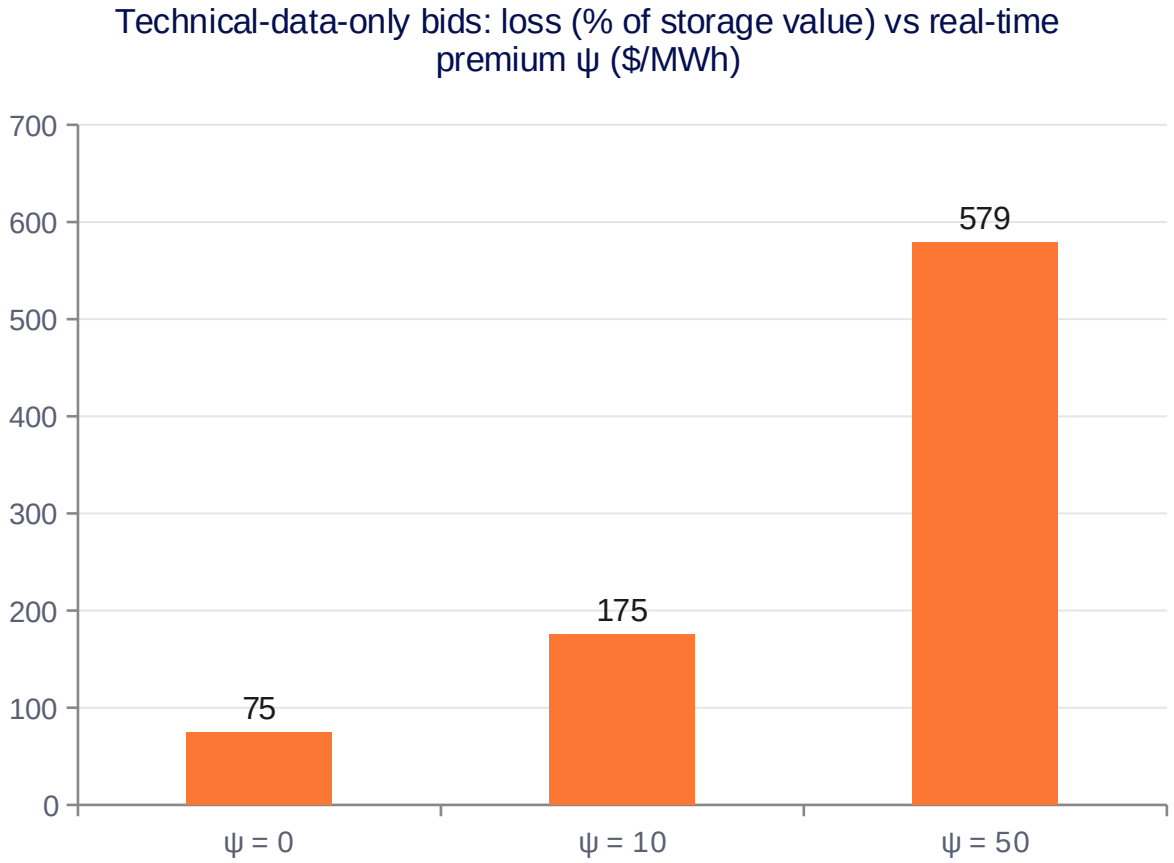
beyond ~20%: worse than no battery

Persistent errors change how much stored energy seems worth: they cost about twice as much.

Result 3: where the loss comes from, and when a forecast is worth having

5 · WHAT FORECAST ERRORS COST





Value of a forecast (ψ = 10)

A forecast beats technical data only up to σ* ≈ 22% for persistent errors, and over the whole tested range for noise. σ* depends strongly on ψ, so ψ is a parameter of the experiments.

Report quantity and price losses separately, and treat the real-time cost ψ as a parameter.

Hypotheses so far, and the plan to finish the experiments

5 · WHAT FORECAST ERRORS COST

	Hypothesis	After the pilot
H1	True prices and quantities give no loss	**confirmed**
H2	Loss grows with σ, from hours where SOC limits bind	grows with σ; split by hour to do
H3	Persistent errors cost more than noise	**yes, about 2× at σ ≥ 10%**
H4	Pre-limit rule beats the solver's duals	mixed; full grid needed
H5	Common errors cost more than independent ones	stage 2
H6	A forecast beats technical data below σ*, growing with ψ	σ* ≈ 22% (ψ = 10, persistent)

Stage 1

Dec 2026

Full grid, one battery: σ × ρ × b × rule × ψ, 500 draws

Stage 2

Jan 2027

5–20 batteries, common share ω, fleet size

Stage 3

Feb 2027

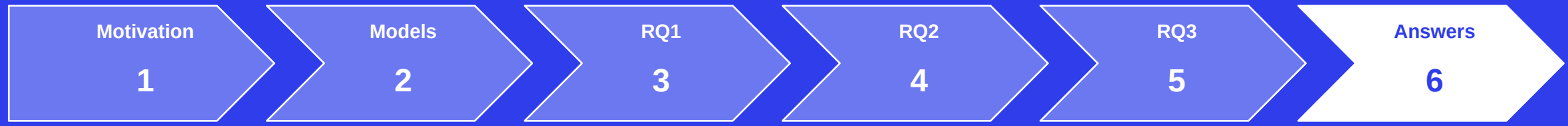
Realistic system, many days, errors calibrated from historical forecasts



[Update as results arrive] The pilot shaped the design; the full grid will size the effects.

PART 6 OF 6

Conclusions



Answers to the research questions

6 · CONCLUSIONS

RQ1

Equivalence

Yes, exactly. Bids at Baldick's thresholds reproduce its dispatch and prices.

Evidence: Proposition 1, Proposition 2; verified on Baldick's 3 cases, zero residuals

RQ2

Information

Only the battery's own parameters and its local prices. Any dual of its own problem works, for any number of batteries.

Evidence: Theorem 1, Proposition 5; 5 batteries, 200 random trials

RQ3

Cost of errors

Preliminary: large, driven by persistent errors and by quantities interval bids cannot carry.

Evidence: pilot, 150 draws: 11.1% lost with perfect prices; 36.5–65.6% with 2–11% noise

The difference between the two designs is information about prices, not the design itself.

What it means for markets

6 · CONCLUSIONS

The reform has a measurable value

The welfare gap between Baldick's design and today's market with well-built bids is the value of the reform.

An intermediate step for operators

Reference bids $\Phi(\theta, \lambda^{ISO})$ from technical data and the operator's forecast: a competitive benchmark for monitoring, with a natural tolerance band.

Where it matters most

Where batteries often become full or empty: that is where prices stop pinning the bids.

Whichever design is chosen, the equivalence makes the comparison fair and quantifiable.

Limitations and next steps

6 · CONCLUSIONS

Limitations

- Convex market: no unit commitment or binaries
- Price-taking participants: no strategic bidding yet
- Linear end-of-day value w
- Interval bids cannot carry quantity information
- Experiments so far: one synthetic day, one battery

Next steps

- Stage 1–3 experiments (Dec 2026 – Feb 2027)
- Calibrate forecast errors on Danish day-ahead prices
- Current design with SOC enforcement (CAISO/NYISO type)
- Monitoring: reference bids vs CAISO-type default energy bids
- Thesis submission: March 2027

[Update before the defense with the final results]

THANK YOU

Questions

Interval bids reproduce Baldick's SOC-based outcome exactly. They need only the battery's own parameters and its local prices: the gap between the designs is an information gap.

[1] R. Baldick, “A state-of-charge based formulation for storage participation in electricity markets: Technical reference,” arXiv:2606.14993, 2026.

Why the current design is a relaxation of Baldick's

BACKUP

$$f^{\mathrm{SOC}\star} = f^{\mathrm{CD}\star} + K \qquad \text{(constant: value of initial energy, rents, terminal benefit)}$$

Proposition 2

Move the SOC rows into the objective with Baldick's optimal multipliers: the SOC variables drop out, the battery's coefficients become exactly the bids (E), and the remaining constant K is the value of the initial energy, the rents at the SOC limits and the terminal benefit. A CD optimum is SOC-optimal only if some SOC path is feasible and consistent with the rents.

Proposition 3 (any stepped bids)

Bids are equivalent if and only if Baldick's schedule lies on each battery's bid and offer curves at Baldick's prices.

Technical-data-only bids

BACKUP

$$u_t = \alpha\,\gamma^{T-t} w - \epsilon, \qquad c_t = \zeta + \beta\,\gamma^{T-t} w$$

When they are exact

If no SOC limit of the battery binds for the rest of the day, these bids are exactly the equivalent bids. In a current design that enforces the SOC in the clearing (CAISO, NYISO), they are always exact: the market then computes the rents itself.

Quantity rule of the experiments

BACKUP

$$\left(u_t + \delta,\ p_t^{\circ}\right),\ \left(u_t - \delta,\ \bar{p} - p_t^{\circ}\right) \qquad \left(c_t - \delta,\ g_t^{\circ}\right),\ \left(c_t + \delta,\ \bar{g} - g_t^{\circ}\right)$$

Example (pilot, hour 15)

Price 38 $/MWh, plan 21.1 MW: bid (38.01 $/MWh, 21.1 MW) and (37.99 $/MWh, 78.9 MW); the market clears 21.1 MW. Baldick's schedule charges 32.6 MW in that hour: the difference is quantity information.


DTU